AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI

ANNUAL EXAMINATIONS (THEORY) 2024

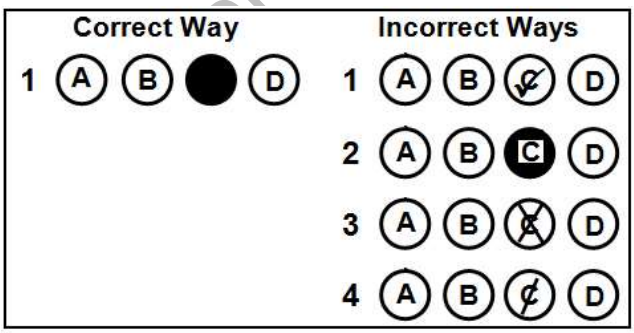
Biology Paper I

Time: 1 hour 30 minutes Marks: 50

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INSTRUCTIONS

1. Read each question carefully.
2. Answer the questions on the separate answer sheet provided. DO NOT write your answers on the question paper.
3. There are 100 answer numbers on the answer sheet. Use answer numbers 1 to 50 only.
4. In each question there are four choices A, B, C, D. Choose ONE. On the answer grid black out the circle for your choice with a pencil as shown below.



Candidate's Signature

5. If you want to change your answer, ERASE the first answer completely with a rubber, before blacking out a new circle.
6. DO NOT write anything in the answer grid. The computer only records what is in the circles.
7. You may use a scientific calculator if you wish.

1. Consider the given research question.

   'How do pollutants impact the ecological functioning and sustainability of a pond?'

   In addition to ecology, this research question encompasses the field of

   A. microbiology.
   B. biotechnology.
   C. marine biology.
   D. freshwater biology.

2. Which of the following shows the CORRECT order of biological levels of organisation from complex to simple?

   A. Spleen → Leucocyte → Mitochondria → Adenine
   B. Mitochondria → Spleen → Leucocyte → Adenine
   C. Mitochondria → Adenine → Leucocyte → Spleen
   D. Spleen → Leucocyte → Adenine → Mitochondria

3. Following are the characteristics of one of the types of polysaccharides.

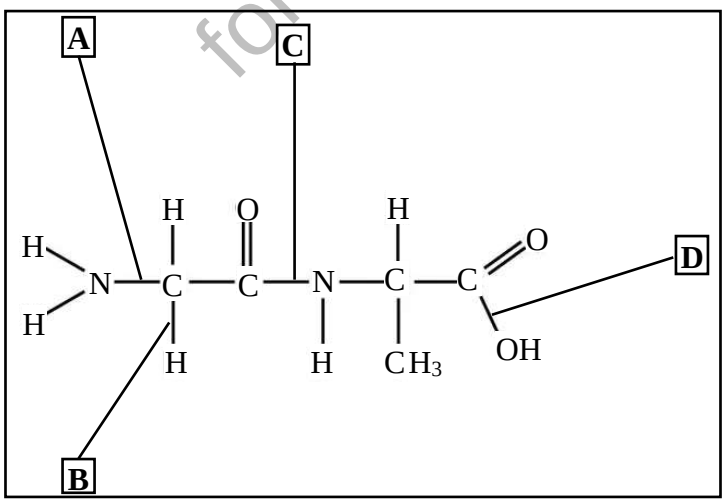
   - High molecular weight
   - Insoluble in water
   - Tasteless

   Based on the given characteristics, this polysaccharide is

   A. lactose.
   B. sucrose.
   C. cellulose.
   D. galactose.

4. The diagram shows the structure of a primary protein.

   Which of the following options indicates peptide linkage (bond) in the given diagram?



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5. In unsaturated lipid molecules, carbon-carbon double bonds are found

   A. within glycerol only.
   B. within fatty acids only.
   C. between fatty acids and glycerol.
   D. within fatty acids and within glycerol.

6. The linkage that is broken by the enzyme amylase is

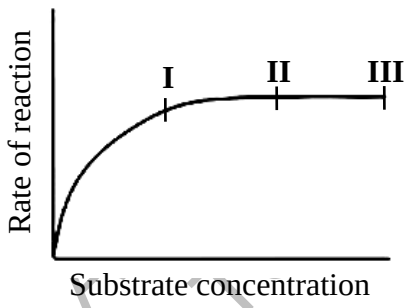
   A. ester.
   B. peptide.
   C. glycosidic.
   D. phosphodiester.

7. The DNA analysis of a fruit fly shows that 22% of nitrogenous bases are thymine.

   The percentage of guanine bases in this DNA would be

   A. 22%.
   B. 28%.
   C. 44%.
   D. 56%.

8. The graph shows the rate of an enzyme-catalysed reaction at increasing substrate concentrations.



   At which of the following point(s) do all active sites of the enzyme seem occupied?

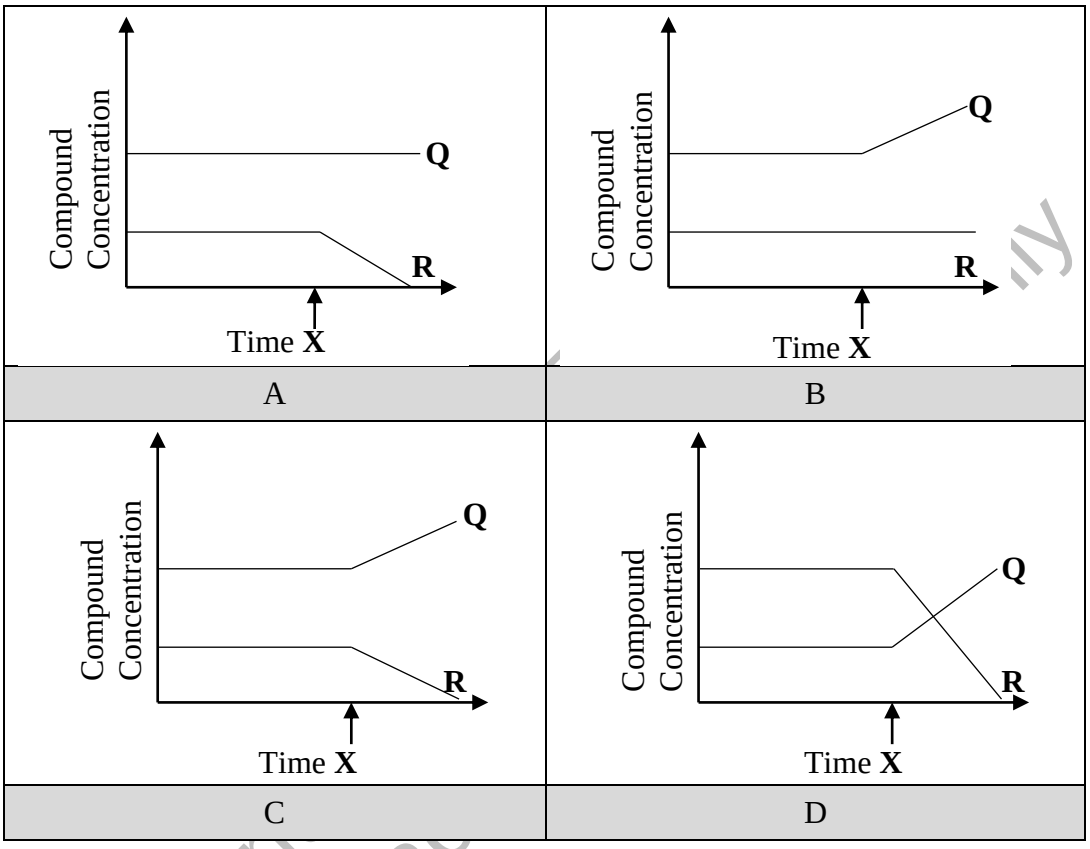
   A. Point I only
   B. Point III only
   C. Points I and II
   D. Points II and III

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9. The given diagram shows an enzyme-controlled metabolic pathway.

Compound **P** $\xrightarrow{\text{Enzyme } \mathbf{I}}$ Compound **Q** $\xrightarrow{\text{Enzyme } \mathbf{II}}$ Compound **R** $\xrightarrow{\text{Enzyme } \mathbf{III}}$ Compound **S**

If enzyme **II** is inhibited at point **X**, then the graph that predicts the resulting concentrations of compounds **Q** and **R** is



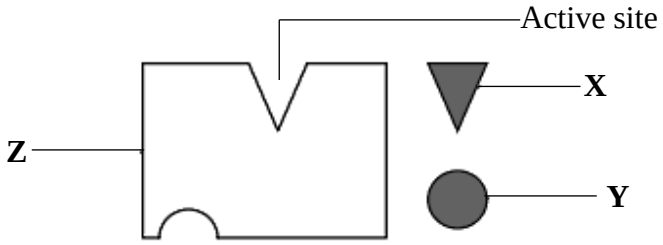
10. The given equation shows a chemical reaction in a human gut.

Amylose / Amylopectin + Water $\xrightarrow[\text{pH 7.0}\ \text{Cl}^-]{\text{Enzyme}}$ Maltose

The identification of enzyme and role of Cl^- in this reaction is

	Enzyme	**Role of Cl^-**
A	amylase	cofactor
B	amylase	coenzyme
C	pectinase	cofactor
D	pectinase	coenzyme

11. The given diagram shows molecules involved in an enzyme-controlled reaction.



The CORRECT identification of **X**, **Y** and **Z** is

	X	**Y**	**Z**
A	substrate	non-competitive inhibitor	enzyme
B	non-competitive inhibitor	enzyme	substrate
C	substrate	competitive inhibitor	enzyme
D	competitive inhibitor	non-competitive inhibitor	substrate

12. The *cis* face of the Golgi apparatus receives materials from the endoplasmic reticulum through transport vesicles; therefore, it is located nearest to the

A. plasma membrane.
B. nuclear membrane.
C. vacuolar membrane.
D. mitochondrial membrane.

13. Which of the following options is CORRECT about chromoplast?

	Membrane-bounded	**Contains Pigment**	**Found in Plants**
A	✓	✓	×
B	×	✓	✓
C	✓	✓	✓
D	✓	×	✓

14. The cell organelle that contains all the necessary enzymes to catalyse the formation of polypeptides by amino acids is

A. ribosome.
B. lysosome.
C. centrosome.
D. peroxisome.

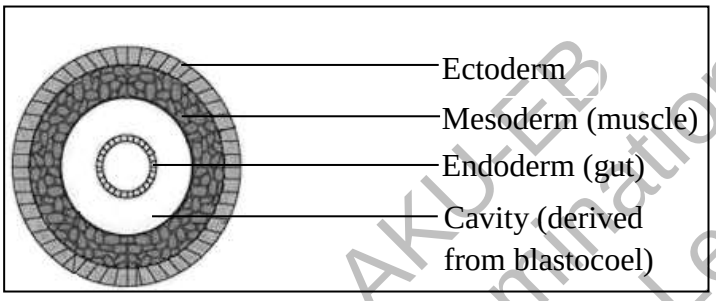
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15. In the process of centrifugation, the cell extract is subjected to centrifugal force which causes the separation of extract into two fractions: a pellet and a supernatant.

The option that identifies the materials in the pellet and supernatant is

	Pellet	Supernatant
A	nuclei	dissolved ions
B	dissolved ions	dissolved ions
C	nuclei	nuclei
D	dissolved ions	nuclei

16. The given diagram represents body tissue organisation of a



A. triploblastic coelomate.
B. diploblastic acoelomate.
C. triploblastic acoelomate.
D. triploblastic pseudocoelomate.

17. The invertebrate which has a closed circulatory system is

A. flat worm.
B. pin worm.
C. earthworm.
D. hookworm.

18. An example of a free-living, non-parasitic flat worm is

A. planaria.
B. tapeworm.
C. liver fluke.
D. blood fluke.

19. Which of the following inhabitants of oceans is considered as the “rain forest of oceans”?

A. Hydra
B. Sponges
C. Coral reefs
D. Colonies of jellyfish

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20. Prions are infectious proteins known to cause

A. athlete's foot.
B. herpes simplex.
C. mad cow disease.
D. mumps and measles.

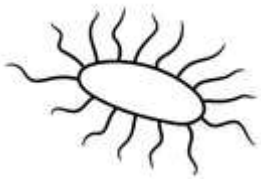
21. Given is the classification of human and chimpanzee.

Human	Chimpanzee
Animalia	Animalia
Vertebrata	Vertebrata
Mammalia	Mammalia
Primate	Primate
Hominidae	Hominidae
Homo	Pan
sapiens	troglodites

The taxonomic categories in which humans and chimpanzee differ are

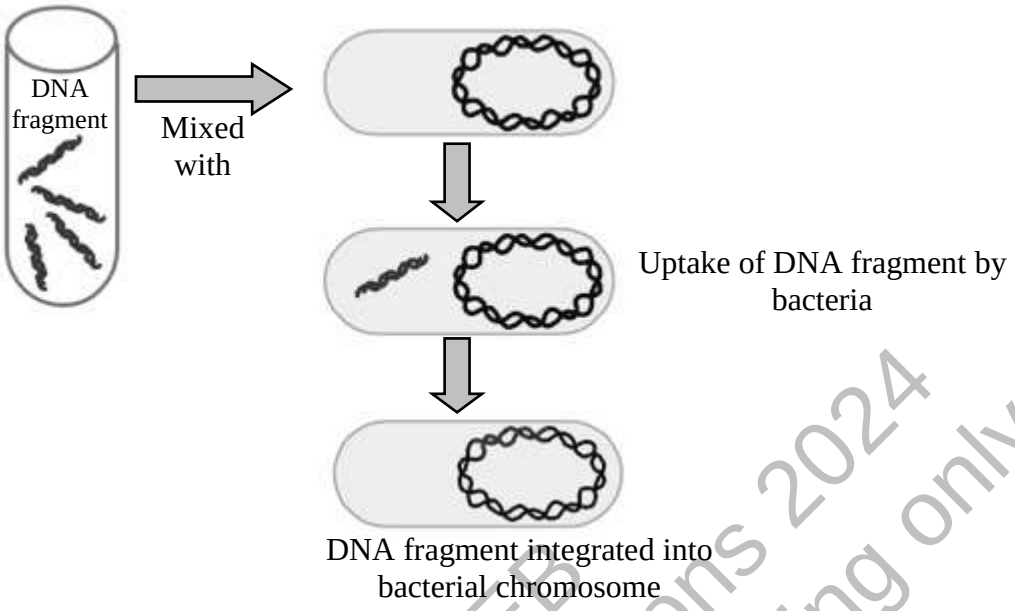
A. order and family.
B. family and genus.
C. order and species.
D. genus and species.

22. Based on the pattern of attachment of flagella, amphitrichous bacterium is identified as

			
A	B	C	D

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23. The given diagram shows a phenomenon in bacteria.



The given phenomenon is identified as

A. conjugation.
B. transduction.
C. transformation.
D. endospore formation.

24. Different methods of sterilisation are used to kill microorganisms.

The method which kills the microorganisms by disrupting their DNA is

A. dry heat.
B. radiation.
C. filtration.
D. moist heat.

25. The shells of foraminiferans contain pores through which cytoplasmic projections are extended.

These cytoplasmic projections help foraminiferans to

A. entangle the prey.
B. enhance water beauty.
C. form limestone deposits.
D. exchange genetic material.

26. Protists are considered as a paraphyletic group because they probably originated from

A. a common moneran ancestor.
B. more than one moneran ancestors.
C. the heritable genetic changes in viruses.
D. the organic substances of the environment.

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27. Deuteromycetes differ from other types of fungi based on the

A. absence of chitin.
B. presence of conidia.
C. absence of sexual reproduction.
D. presence of absorptive nutrition.

28. The option that identifies the stored form of food in fungi and plants is

	Fungi	**Plant**
A	starch	glycogen
B	starch	starch
C	glycogen	glycogen
D	glycogen	starch

29. The fungus that possesses non-septate and multinucleate hyphae

A. yeast.
B. rhizopus.
C. mushroom.
D. penicillium.

30. The structure of a plant that characterises a monocot plant is

			
A	B	C	D

31. The FIRST group of plants that formed true leaves and roots was

A. Psilopsida.
B. Lycopsida.
C. Pteropsida.
D. Sphenopsida.

32. In moss plant, the parts that constitute the gametophyte are located inside the

A. leaves and seta.
B. seta and capsule.
C. shoots and leaves.
D. rhizoids and capsule.

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33. The advanced characteristics of the sporophyte of hornworts (anthocerotae) that enable them to thrive better on land are the presence of

A. rhizoids which help to anchor the plant.
B. stomata and chloroplasts in the epidermis.
C. horn-like projections out of the gametophyte.
D. mature sporangium that splits open at the ends.

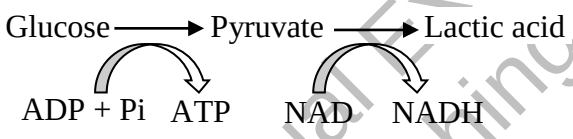
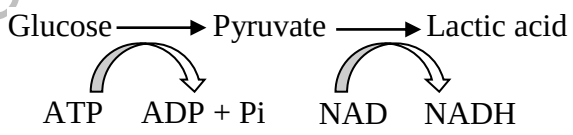
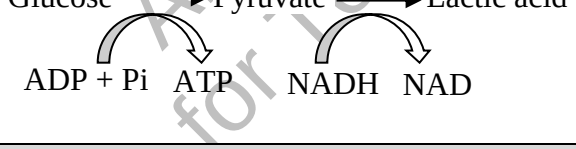
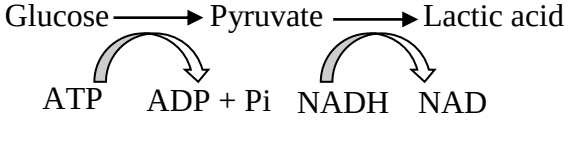
34. The cells of the embryo sac of an apple plant include

A. antipodal cells, polar nuclei, sperm cell and tube cells.
B. synergids, egg cell, polar nuclei and antipodal cells.
C. synergids, egg cell and generative cells.
D. tube cells and sperm cell.

35. During photorespiration, ribulose biphosphate carboxylase combines with oxygen gas which results in lowering the overall rate of

A. ADP synthesis.
B. NAD formation.
C. nitrogen fixation.
D. carbon dioxide fixation.

36. The equation that summarises the conversion of glucose into lactic acid during anaerobic respiration is

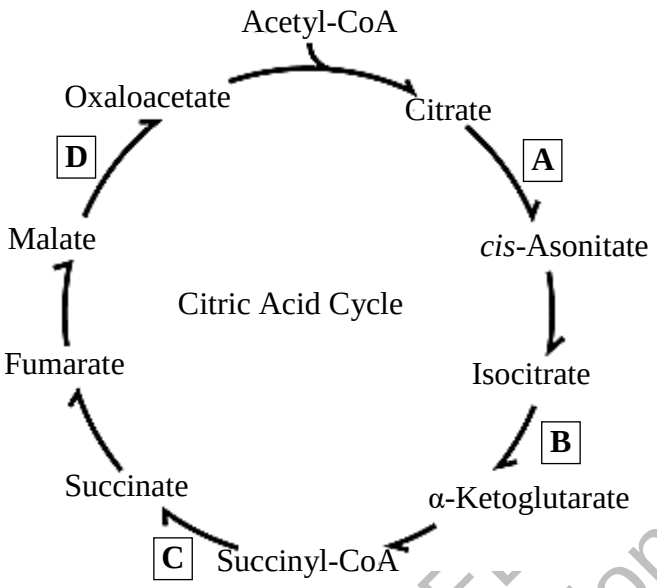
	
A	B
	
C	D

37. In light dependent reactions of photosynthesis, the cyclic electron transport yields only

A. ATP.
B. water.
C. oxygen.
D. NADPH.

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38. In the given Krebs cycle (citric acid cycle) of cellular respiration, the step where ATP is produced is labelled as



39. In the human digestive system, chyme that enters the duodenum, is a mixture of partially digested food and

A. bile.
B. gastric juice.
C. intestinal juice.
D. pancreatic juice.

40. The animal which has a gut with single opening is

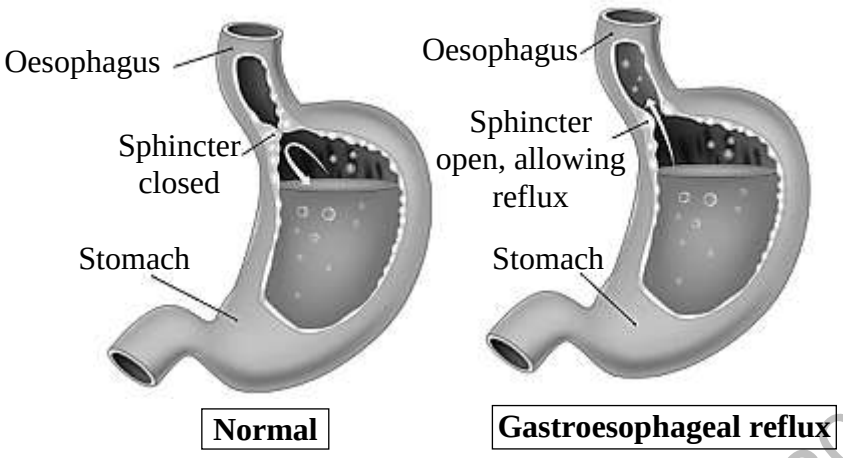
A. planaria.
B. cockroach.
C. earthworm.
D. roundworm.

41. The pathway by which lipoproteins (products of lipid digestion) are absorbed from the intestinal villi into the blood is

A. epithelial cells → interstitial fluid → capillaries → subclavian vein.
B. epithelial cells → capillaries → interstitial fluid → thoracic vein.
C. lacteal → lymph → thoracic lymph duct → subclavian vein.
D. lacteal → interstitial fluid → capillaries → thoracic vein.

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42. Consider the given diagram.



The condition shown in the case of gastroesophageal reflux causes

A. diarrhoea.
B. dyspepsia.
C. haemorrhoids.
D. food poisoning.

43. In the human respiratory tract, temperature and humidity of the inhaled air increases in the

A. trachea.
B. bronchi.
C. nasal cavity.
D. alveolar space.

44. During breathing in humans, the contraction of external intercostal muscles causes

A. compression of the rib cage.
B. outward movement of ribs.
C. inward movement of sternum.
D. relaxation of the diaphragm muscles.

45. The capacity of oxyhaemoglobin to release oxygen at tissue level is MAINLY dependent on

A. high concentration of oxygen in blood.
B. lower pressure of nitrogen in atmosphere.
C. higher pressure of carbon dioxide in blood.
D. low temperature of body in cold environment.

46. The heart of a fish acts as ‘single circuit’ where the blood from gills flows to

A. atrium.
B. dorsal aorta.
C. ventral aorta.
D. sinus venosus.

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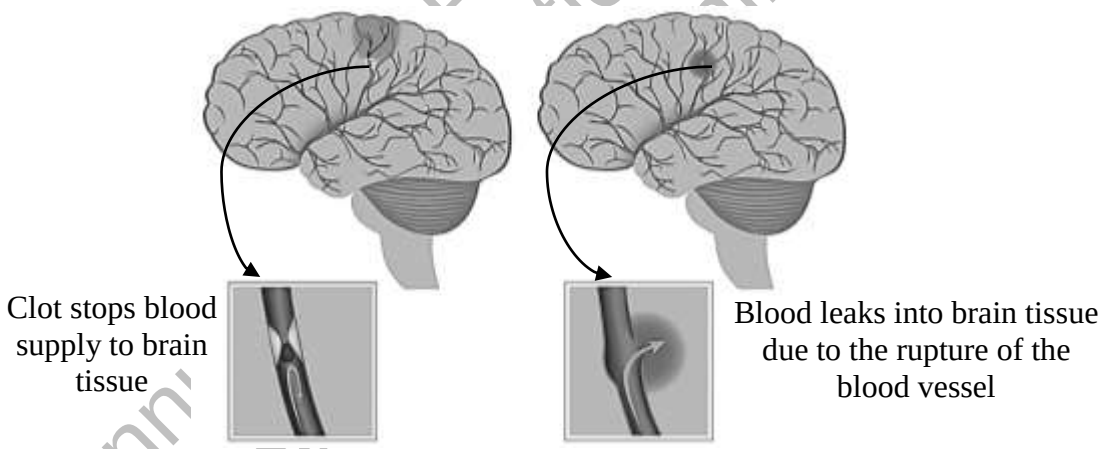
47. The set of factors which causes the SLOWEST rate of transpiration in terrestrial plants is

	Humidity	Temperature	Wind Speed
A	high	low	high
B	low	high	high
C	high	low	low
D	low	high	low

48. In the human body, the lymphoid masses that produce lymphocytes are

A. liver and gall bladder.
B. spleen and thymus gland.
C. thyroid and parathyroid glands.
D. lymph vessels and bone marrow.

49. Consider the given diagrams **X** and **Y**.



Based on the information provided, conditions **X** and **Y** are identified as

	X	Y
A	stroke	haemorrhage
B	haemorrhage	stroke
C	haemorrhage	arteriosclerosis
D	stroke	atherosclerosis

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50. During an epidemic, an immunisation programme is organised to provide immediate protection to the vulnerable people.

The type of immunity provided to these people would be

A. active and natural.
B. active and artificial.
C. passive and natural.
D. passive and artificial.

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